

PH.D. APPLIED QUALIFYING EXAM 2022 (TAKE HOME)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. The exam consists of three problems, all of which you need to answer. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have questions, please email Prof. Han Xiao (hxiao@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

- (1) Type up your answers to the two problems in two separate Word or Latex files.
- (2) Create a folder for each problem. The folder must contain the following:
 - (a) A word or pdf file containing your answers to the problem, including all tables, figures and Appendices.
 - (b) An R (or python) file containing the codes used to generate your results.
 - (c) A word or a text file that gives a summary of the codes submitted.
- (3) Place the two folders into a single folder titled **yourlastname_yourfirstname_Appliedexam2022**. Zip this folder and email it to hxiao@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 22, Friday.

Problem 1

Suppose that $n \times p$ covariate matrix $X = (x_1^T, \dots, x_n^T)^T$ and $n \times 1$ response vector $Y = (y_1, \dots, y_n)^T$ are generated as follows: for $i = 1, \dots, n$,

$$x_i = Wz_i + u_i,$$

$$y_i = \theta^T z_i + v_i,$$

where W is a $p \times d$ constant matrix, θ is a $d \times 1$ constant vector, $z_1, \dots, z_n$ are independent with $z_i \sim N(0, I_d)$, $u_1, \dots, u_n$ are independent with $u_i \sim N(0, I_p)$, $v_1, \dots, v_n$ are independent with $v_i \sim N(0, \sigma_v^2)$, z_i 's, u_i 's, and v_i 's are also independent of each other, and I_d, I_p denotes the $d \times d$ and $p \times p$ identity matrix respectively.

We are interested in studying the in-sample and out-of-sample prediction errors for the ridge estimator of β , which is defined as

$$\hat{\beta}_\lambda = \operatorname{argmin}_\beta \left\{ \frac{1}{n} \sum_{i=1}^n (y_i - \beta^T x_i)^2 + \lambda \|\beta\|_2^2 \right\},$$

where $\lambda \geq 0$ is a tuning parameter. The in-sample and out-of-sample prediction errors are defined as

$$\text{Err.in} = \frac{1}{n} \sum_{i=1}^n (x_i^T \hat{\beta}_\lambda - x_i^T \beta)^2,$$

$$\text{Err.out} = E_{\text{out}}(x_0^T \hat{\beta}_\lambda - x_0^T \beta)^2,$$

where x_0 is a test covariate vector, independent of the training data (X, Y) , and $E_{\text{out}}(\cdot)$ is taken with respect to x_0 , *conditionally on the training data* (X, Y) .

(i) Show that (x_i, y_i) , $i = 1, \dots, n$, satisfy a linear model:

$$y_i = x_i^T \beta + \varepsilon_i,$$

where $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$. Show that

$$\beta = W(I_d + W^T W)^{-1} \theta, \quad \sigma_\varepsilon^2 = \sigma_v^2 + \theta^T (I_d + W^T W)^{-1} \theta$$

(ii) Show that $x_i \sim N(0, \Sigma)$ for $i = 1, \dots, n$, where

$$\Sigma = I_p + W W^T$$

(iii) Find the expectation of Err.in, called expected Err.in, over the distribution of Y conditionally on X . Express your answer in terms of $\beta, \sigma_\varepsilon^2$, the (nonzero) eigenvalues and their corresponding eigenvectors of $\hat{\Sigma} = \frac{X^T X}{n}$.

In the remaining questions, suppose that (W, θ, σ_v^2) are specified as follows with $p = 2^d$:

$$W = (w_1^T, \dots, w_p^T)^T / \sqrt{d}, \quad \theta = \mathbf{1}_d / \sqrt{d}, \quad \sigma_v^2 = 1,$$

where $w_1, \dots, w_p$ (with $p = 2^d$) correspond to all possible length- d arrangements of -1 or 1 , and $\mathbf{1}_d$ denotes a $d \times 1$ vector of all ones. For example, for $d = 3$ and $p = 8$,

$$\begin{pmatrix} w_1^T \\ \vdots \\ w_p^T \end{pmatrix} = \begin{pmatrix} -1 & -1 & -1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

(iv) For the preceding specification, show that the diagonal elements of Σ are all equal to 2, i.e., the marginal variances of the components of x_i are 2. Moreover, show that the eigenvalues of Σ , in the decreasing order, are

$$\begin{aligned} \lambda_1 = \dots = \lambda_d &= \frac{p}{d} + 1, \\ \lambda_{d+1} = \dots = \lambda_{2^d} &= 1. \end{aligned}$$

For a simulation study, generate training and test data as follows in R (please run these codes **exactly with the random seed**):

```
#-----
set.seed(1)

### training data
d <- 9
n <- 50
p <- 2^d

theta <- rep(1/sqrt(d), d)
P <- expand.grid(replicate(d, 0:1, simplify = FALSE))
W <- as.matrix((2*P - 1)/sqrt(d))

Z <- matrix(rnorm(n*d), nrow = n, ncol = d)
Y <- Z %*% theta + rnorm(n)
X <- Z %*% t(W) + matrix(rnorm(n*p), nrow = n, ncol = p)
#-----
```

For all questions requiring coding, please provide both your R codes and output.

(v) For the following choices of λ ,

$$10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 2.5, 5, 7.5, 10, 20, 30, 40, 50, 100, 250, 500, 10^3, 10^4, 10^5, 10^6$$

compute and *report* Err.in and Err.out and plot them against $-\log(\lambda)$, given the training data generated above.

(vi) Repeat the experiment for 100 times, i.e., generating training data in the same way as above (except for removing the set.seed line) and computing Err.in and Err.out. Plot the averages of Err.in and Err.out against $-\log(\lambda)$.

(vii) Given the training data as used in Question (v), compute and *report* the expected Err.in using your formula from Question (iii) and plot them against $-\log(\lambda)$ together with Err.in from Question (v). Moreover, please repeat the experiment for 100 times (except for removing the set.seed line), plot the averages of the **expected** Err.in against $-\log(\lambda)$ together with the averages of Err.in from Question (vi).

Problem 2

Consider the following linear model with random effect u_i :

$$y_{ij} = \beta_0 + u_i + x_{ij}\beta + \epsilon_{ij},$$

where $i = 1, \dots, k$ represents each of k groups and $j = 1, \dots, n_i$ represents the number of observations in the i -th group, u_i are i.i.d. samples from $N(0, \sigma_u^2)$, ϵ_{ij} are i.i.d. samples from $N(0, \sigma_e^2)$, u_i and ϵ_{ij} are independent, and σ_u and σ_e are known. Define the unknown parameters by $\theta = (\beta_0, \beta)$. The observed data consists of (y_{ij}, x_{ij}) pairs while u_i 's are unobserved.

- (1) To estimate the parameters by EM algorithm, we assume the complete data to be (y_i, u_i) , where $y_i = (y_{i1}, \dots, y_{in_i})$. Write down the log-likelihood for the complete data.
- (2) Assuming that σ_u and σ_e are known, starting with some initial values denoted by $\beta_0^{(0)}$ and $\beta^{(0)}$, write down an explicit updating procedure based on the EM-algorithm, explaining the E -Step and the M -Step clearly.
- (3) Let $k = 30$, $n_1 = \dots = n_{30} = 20$, $\sigma_u^2 = 1$, $\sigma_e^2 = 1.5$, $\beta_0 = 0.5$, $\beta = 1.5$, x_{ij} are random samples from $\text{Unif}(10, 60)$. Simulate one set of (y_{ij}, x_{ij}) and write your own code to estimate θ based on the answer in (3). Repeat this procedure 100 times, evaluate the estimation performance based on mean squared errors (MSE) of the estimated parameters.

Problem 3

Importance sampling (IS) is a technique that can be used to approximate the expectation of a function with respect to an intractable distribution. We assume that the random variable we want to compute the expectation is of the form $f(x)$, where the distribution of x is absolutely continuous and $p(x)$ is the density. The goal is to compute

$$\mu = E(f(x)) = \int f(x)p(x)dx.$$

Define $q(x)$ to be the proposal distribution. By sampling $x_1, \dots, x_n$ from $q(x)$, important sampling algorithm estimates μ by

$$\hat{\mu}_q = \frac{1}{n} \sum_{i=1}^n \frac{f(x_i)p(x_i)}{q(x_i)},$$

assuming $f(x)p(x)/q(x)$ is computable.

- (1) Prove that $\hat{\mu}_q$ is an unbiased estimator of μ , i.e., $E_q(\hat{\mu}_q) = \mu$, where $E_q(\cdot)$ represents expectation with respect to the proposal distribution q .
- (2) What is the variance of $\hat{\mu}_q$?
- (3) What is the optimal choice of the proposal distribution, $q(x)$, that minimizes the variance in (2)?
- (4) Use the idea of importance sampling with the proposal distribution $q \sim N(6, 1)$ to estimate $P(y \geq 3)$, where y follows a standard normal distribution, $N(0, 1)$. Write your own importance sampling algorithm and report the estimated mean and variance of $P(y \geq 3)$ based on 1000 replicates of simulations with size 100.
- (5) Based on the answer in (3), can you recommend a better proposal in compared to $N(6, 1)$ in (4) ?